Unfold

Learn Apple, one fold at a time.

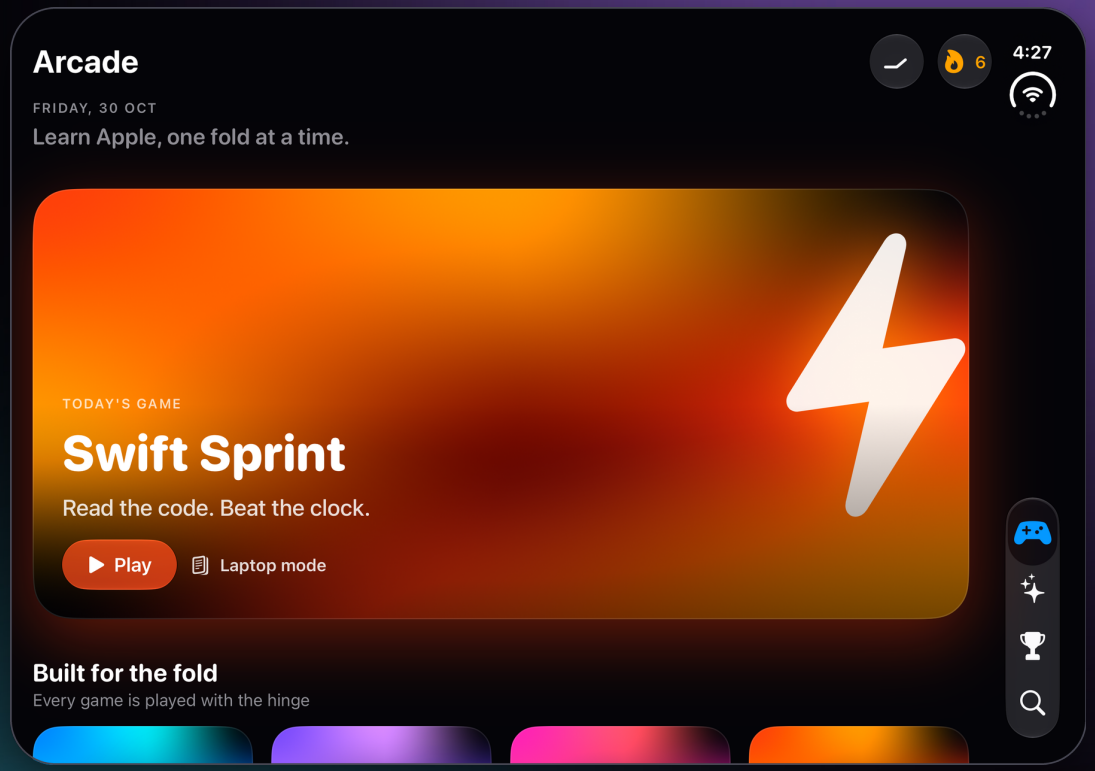
An AI-powered learning arcade built for iPhone Duo and iPhone 18 Pro Max.

4 fold-native games

On-device Apple Intelligence

Liquid Glass

iOS 27.1 hinge APIs



Running on iPhone Duo's 7.6-inch inner display

Sofiyan Shaikh · October 2026

THE IDEA

Apple Arcade polish, with a curriculum inside



Play to learn

Four mini-games teach Swift, Apple frameworks, concurrency and 50 years of Apple history. Every mistake explains itself.



The fold is the controller

On iPhone Duo the hinge is the input: flip flashcards, dial through history, guard Swift's isolation boundary.



A private AI coach

Apple Intelligence runs on the iPhone, reviews every round and answers questions grounded in Unfold's curriculum.

4
fold-native games

80
Apple concepts

100%
on-device AI

0
third-party libraries

Built on Apple's real foldable APIs, not a fake slider

```
.onHingeChange { _, context in
    fold.hingeChanged(degrees: context.hinge?.angle.degrees)
}

geo.reservedRegions(kind: .division)  // the crease
```

Live hinge angle and status

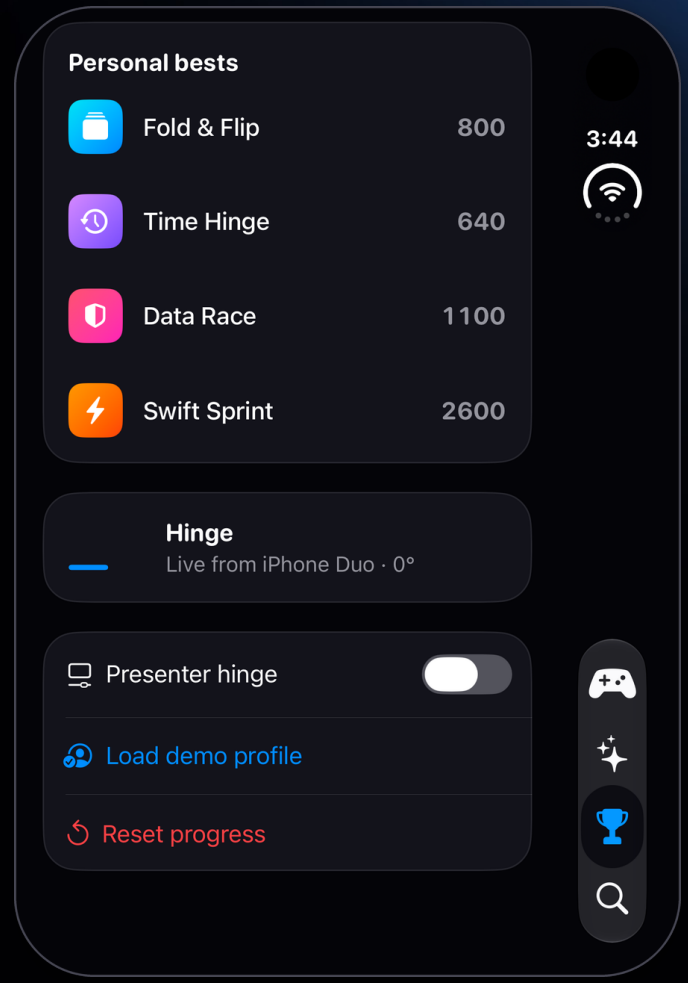
onHingeChange feeds one FoldState; the iPhone Duo Simulator reports it live (right).

Displacement on the crease

The fold is a reserved region, so content moves off it instead of being cut in half.

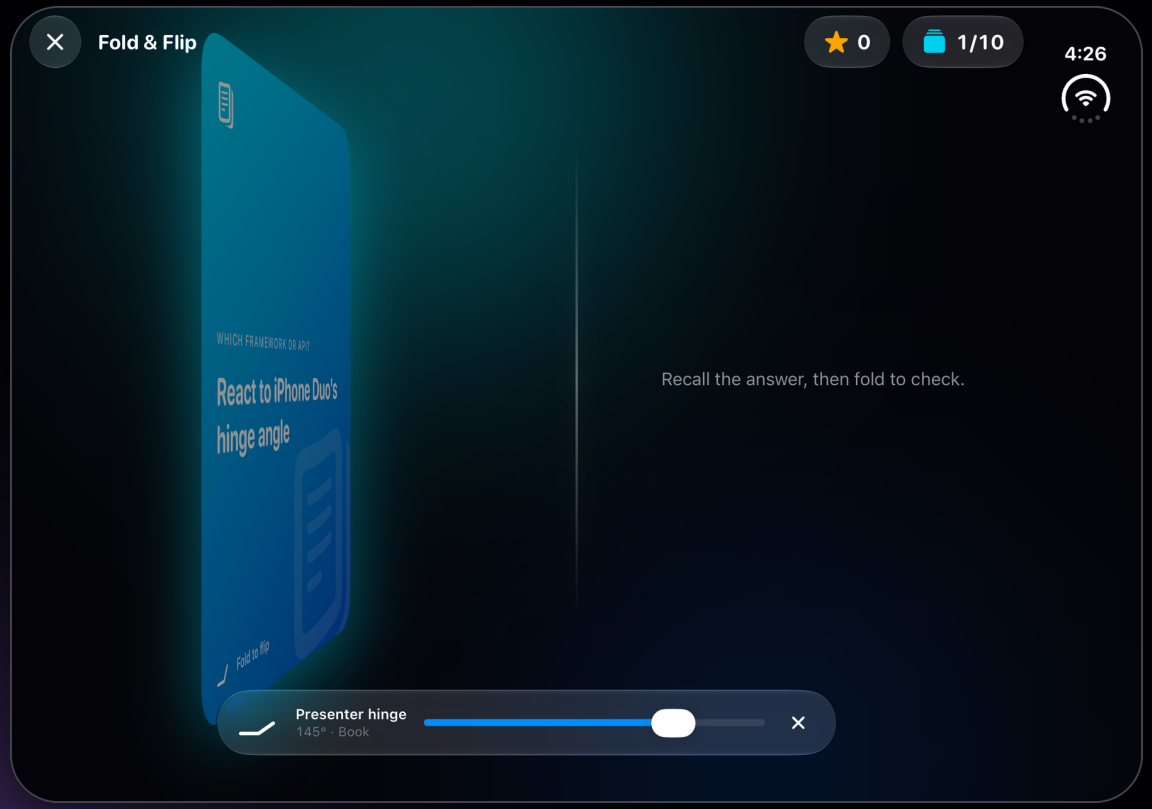
Presenter hinge

A Liquid Glass slider drives the same engine, so fold play can be shown on any iPhone.

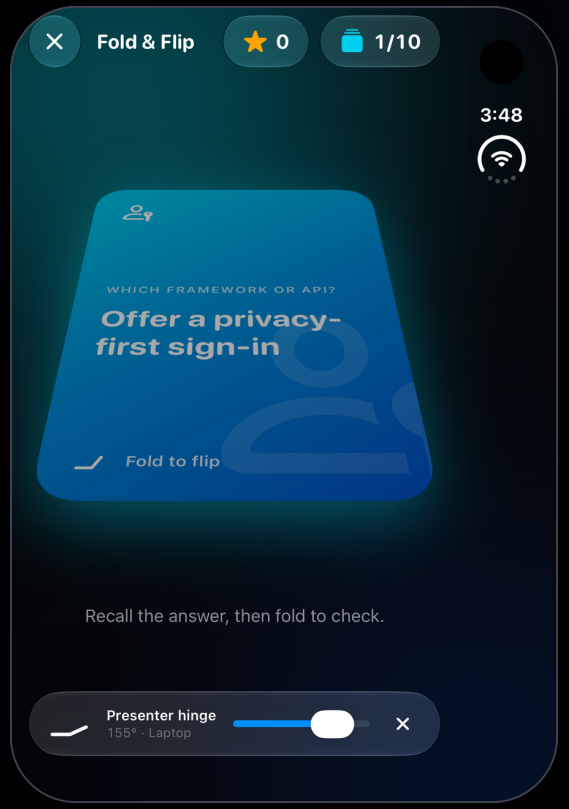


GAME 1 · FRAMEWORKS

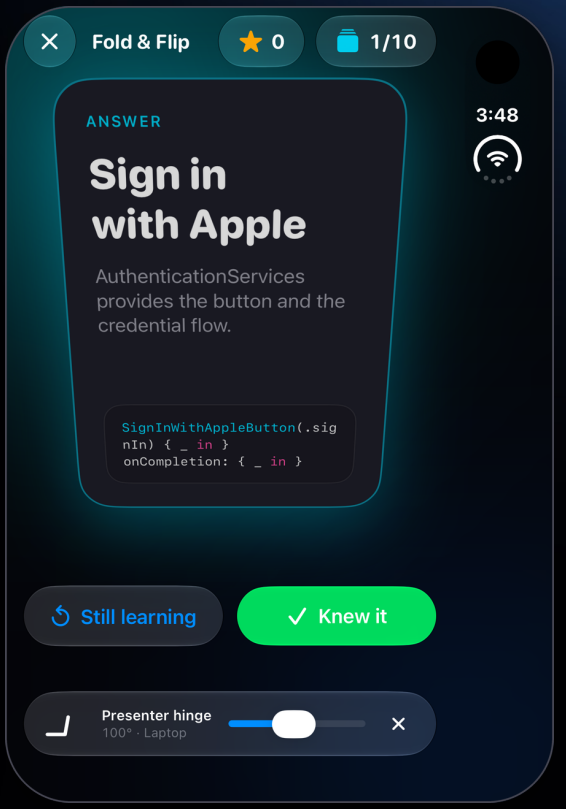
Fold & Flip: the card turns on the hinge's own axis



Book pose · the card turns like a page



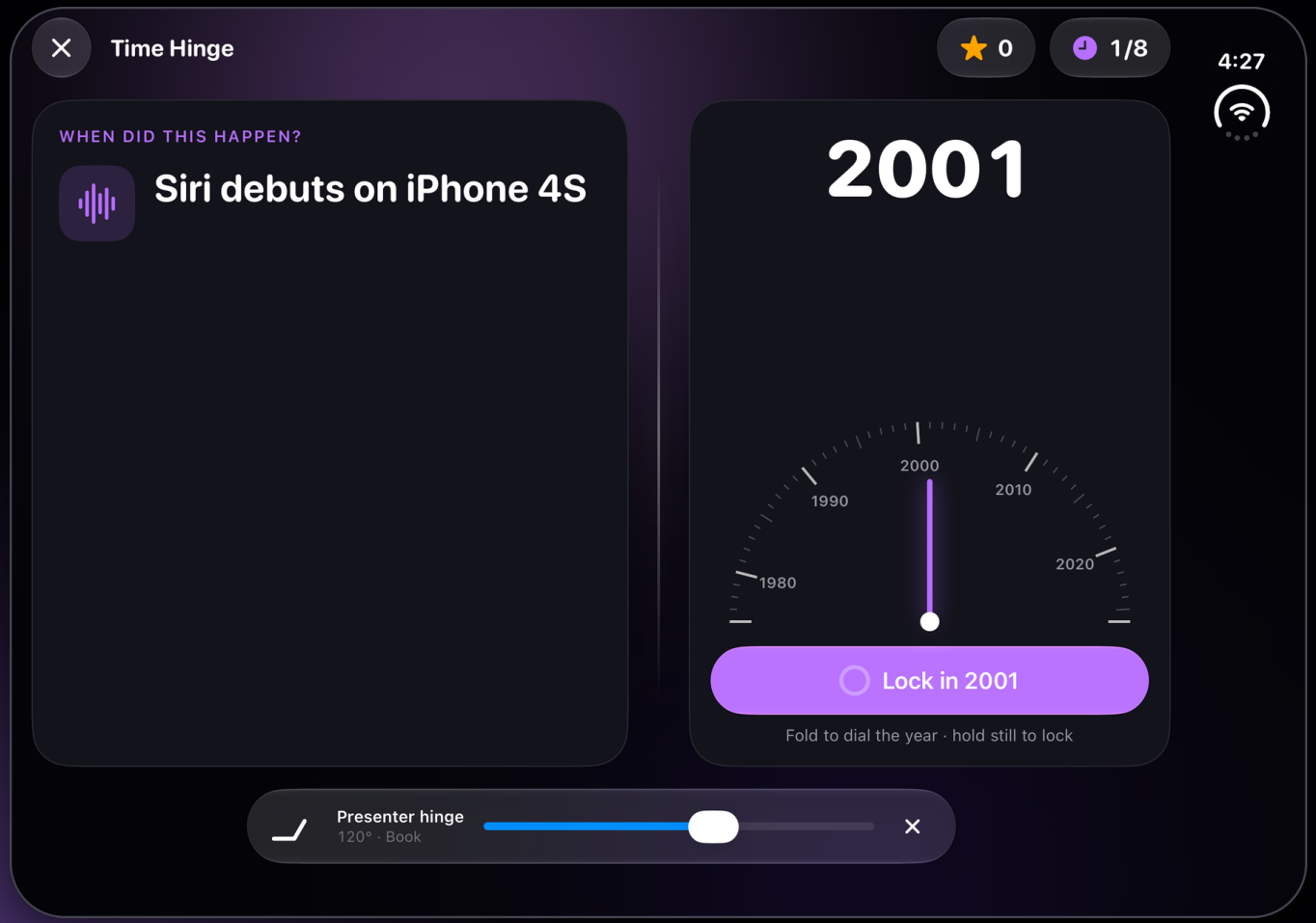
Laptop pose, 155° · tilts with the hinge



Past 90° · the answer, then unfold to deal

GAME 2 · ECOSYSTEM

Time Hinge: the hinge is a dial through fifty years of Apple



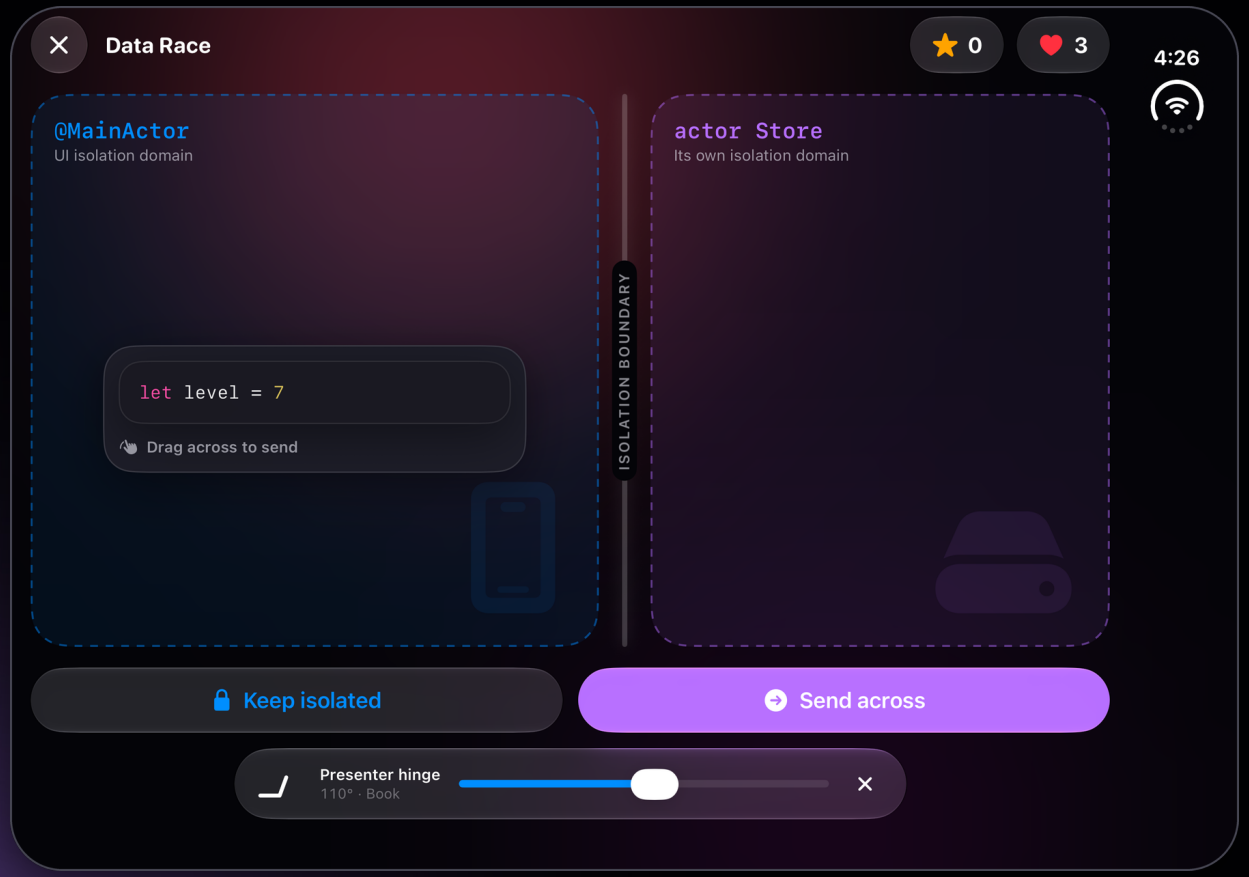
1 **Read the moment**
From Apple's founding in 1976 to iPhone Duo in 2026.

2 **Fold to dial the year**
The hinge's 40°–180° sweep maps onto 1976–2026, like a protractor.

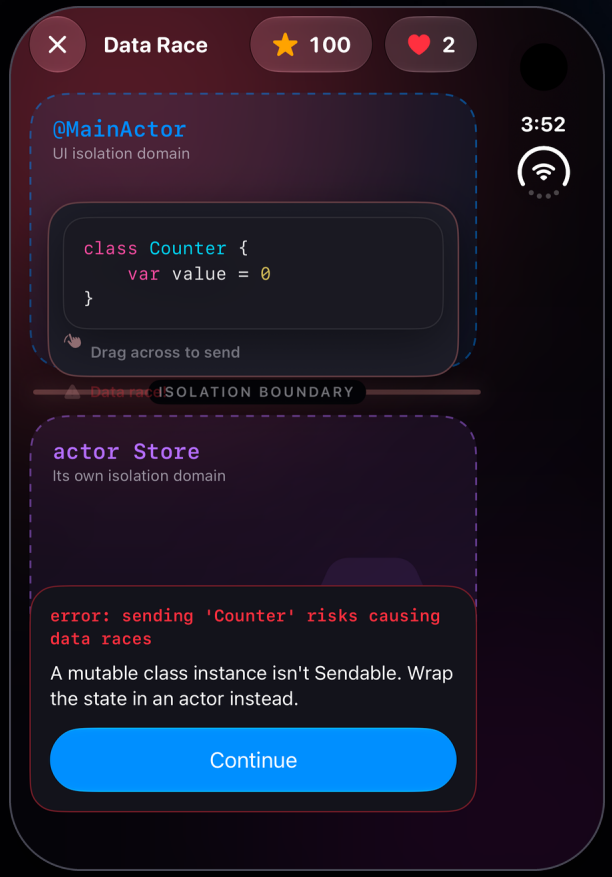
3 **Hold still to lock**
After 1.3 seconds it locks. Exact scores 100; one year off scores 70.

GAME 3 · SWIFT CONCURRENCY

Data Race: the crease becomes Swift's isolation boundary



Book pose · the boundary sits exactly on the crease



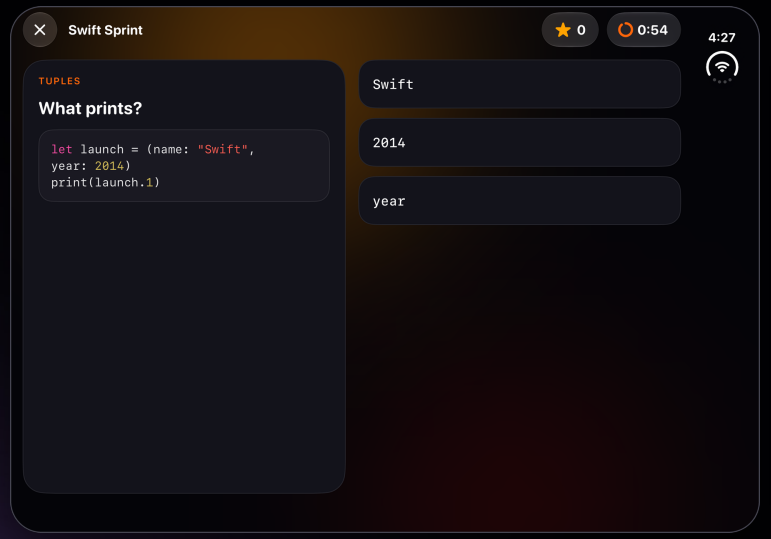
Cover · a wrong move

Only Sendable crosses
Drag values from @MainActor to actor Store, or keep them home.

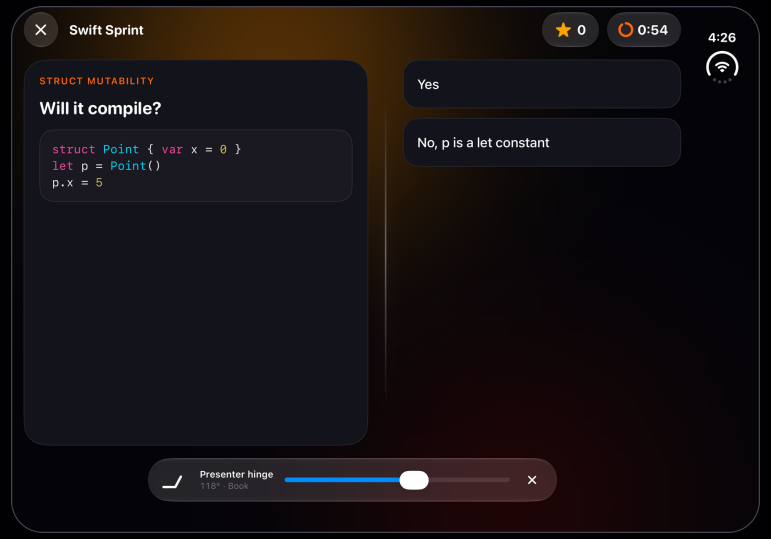
Real diagnostics
error: sending 'Counter' risks causing data races

Pro-level traps
@MainActor classes are Sendable; final immutable classes are not.

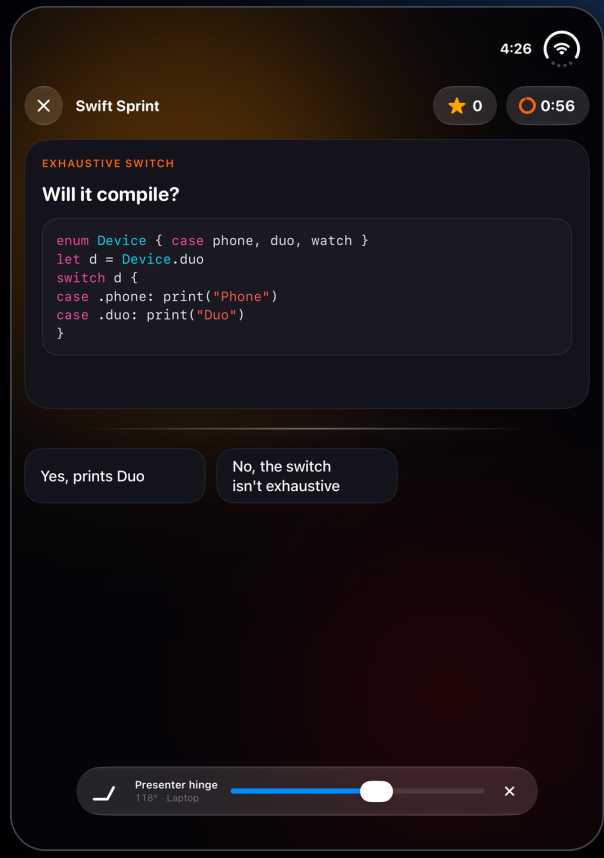
One layout, moved and never redesigned, in every pose



Flat · side by side



Book · split on a vertical crease



Laptop · answers on the resting half

APPLE INTELLIGENCE · ON DEVICE

A private AI coach that knows what you got wrong

Your question or your mistakes → Retrieve the closest curriculum notes → **FoundationModels, on device** → Streamed answer and debrief

Grounded in Unfold's curriculum

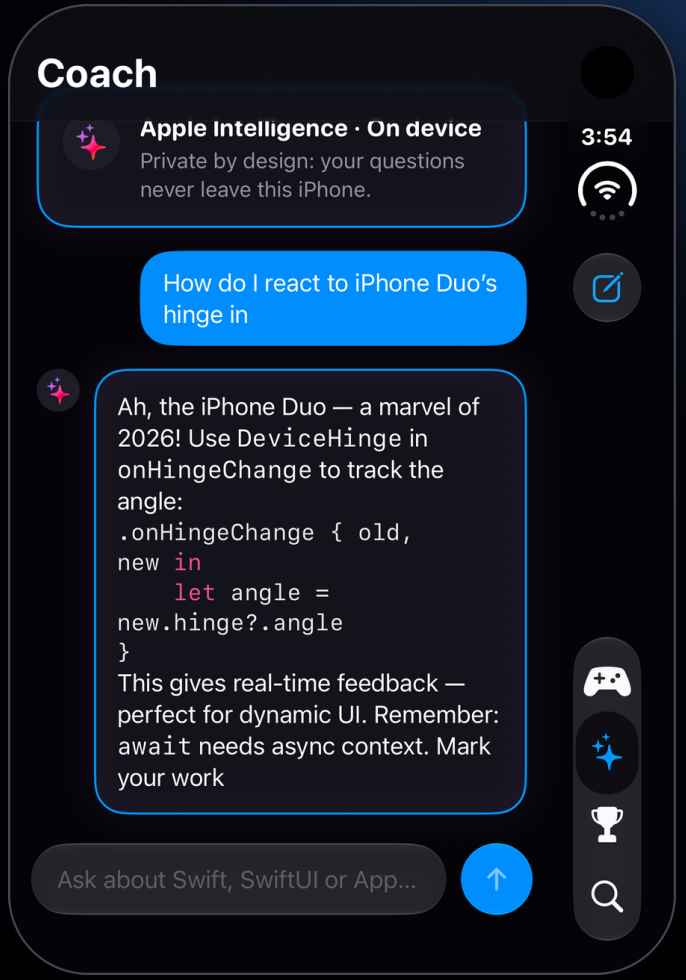
The on-device model can explain APIs newer than its training, like onHingeChange.

A debrief after every round

Three sentences written from your actual mistakes: praise, the idea to fix, one tip.

Private and resilient

Nothing leaves the iPhone. Without Apple Intelligence, Coach answers offline from the same notes.



Liquid Glass only where Apple says it belongs

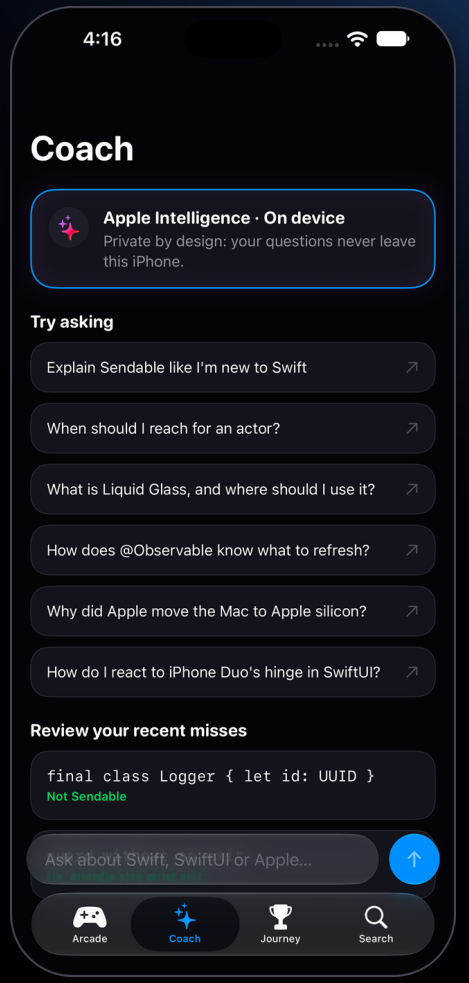
> *“A distinct functional layer for controls and navigation elements.”*
>
> Apple, Adopting Liquid Glass

Glass on

- Tab bar, which iOS moves to the right edge of the cover display
- Game HUD in a GlassEffectContainer
- Coach composer and the presenter hinge
- .glass and .glassProminent buttons

Never

- On content cards: content stays solid
- Glass on glass
- Custom backgrounds behind system bars
- Effects that ignore Reduce Motion



Every colour in Unfold means exactly one thing

Blue
Only things you can tap

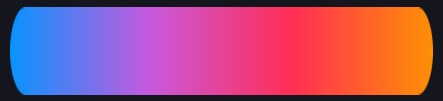
Orange
The Swift world: momentum

Cyan
Frameworks: calm building

Purple
Ecosystem: heritage

Green + ✓
Correct

Red + ✕
Wrong, for a moment only



Intelligence gradient only when Apple Intelligence is speaking

Every colour is paired with a symbol, so colour is never the only signal.

What the research says

Red primes avoidance; blue primes approach.
Mehta & Zhu, Science, 2009

Brief red before a test lowered scores.
Elliot et al., J. Exp. Psych.: General, 2007

So red only flashes, then the explanation takes over. Dark-only, like Apple Fitness: P3 colour reads best on OLED black.

LEARNING DESIGN

Built on how people actually learn

Retrieval practice
Every game asks before it tells. Recalling builds memory faster than rereading.

Immediate, specific feedback
Each mistake shows the right answer and why, the moment it happens.

Spaced repetition
Leitner boxes bring back the flashcards you missed before the ones you know.

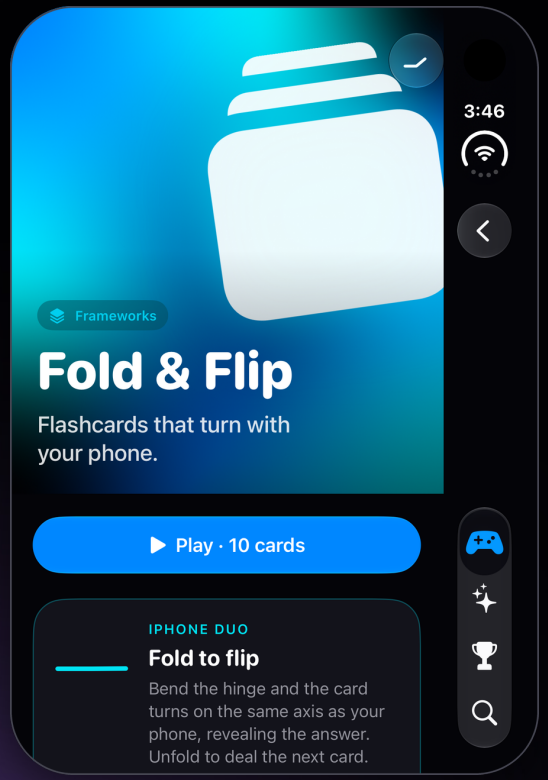
Mastery you can see
Activity-style rings track each world; streaks, XP and personal bests keep you coming back.



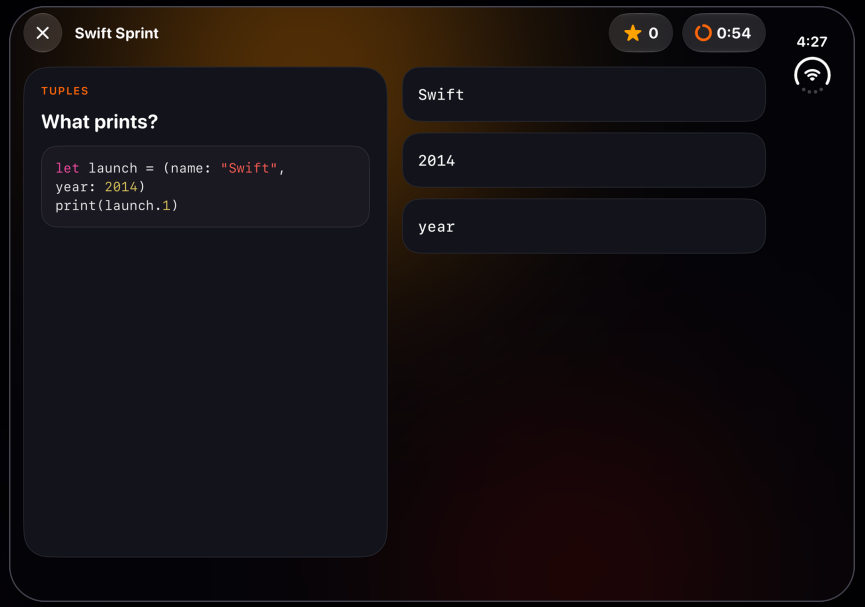
Journey · three mastery rings, one per world

IPHONE DUO + IPHONE 18 PRO MAX

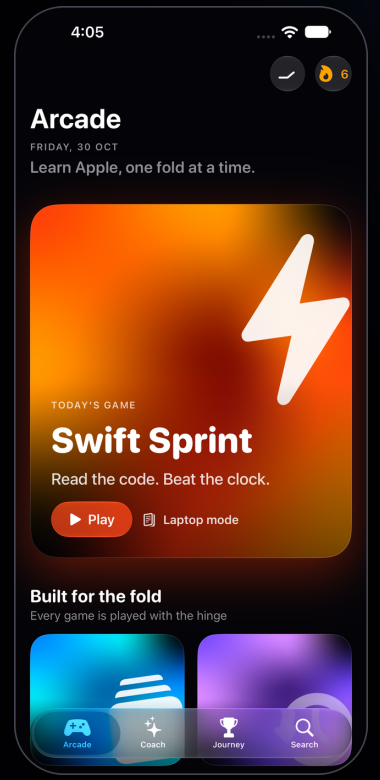
One codebase that adapts to every iPhone



Duo cover · 466 × 678 pt



Duo inner · 951 × 669 pt



18 Pro Max · 440 × 956 pt

- SwiftUI + Observation
- Swift default MainActor isolation
- FoundationModels
- DeviceHinge + ReservedRegion
- GlassEffectContainer
- MeshGradient · zoom transitions
- No third-party code · 0 warnings



Unfold

Learn Apple, one fold at a time.

1
Fold & Flip
Bend the hinge; watch the card turn

2
Data Race
Send a class across: a data race

3
Coach
Ask how to read the hinge

4
Journey
See the mastery rings fill

Live demo · about 3 minutes